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RESEARCH-ARTICLE

Stopped yet Completed: Exploring the Relationships between Session-stopping Reasons, Information Types, and Cognitive Activities in Cross-Session Searches

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Stopped yet Complete: Exploring the Relationships between Session-Stopping Reasons, Information Types, and Cognitive Activities in Cross-Session Searches

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ABSTRACT

The aim of this study is to explore connections among the reasons that lead users to stop a search session, the types of information they found during the session, and the cognitive activities involved in interacting with such information – all in the context of complex tasks involving multiple search sessions. Prior research has primarily examined search-stopping behavior in single search sessions, focusing on instances when users identify needed information or are satisfied with their findings. However, recent insights from the Search as Learning (SAL) community highlight the relationships between user search behavior, the nature of the information they encountered, and various cognitive activities during the information-searching process. We conducted a diary study with 25 participants engaged in real-life tasks that spanned multiple search sessions over time. Our analysis found six predominant reasons users elected to stop a search session during the cross-session search process. We also examined the types of information that participants found and the cognitive activities that they engaged in during search sessions. We found statistically significant associations between the information types found and the session stopping reasons. We did not find significant associations between the cognitive activities and stopping reasons, but did observe that almost half the sessions reached the *evaluate* level of cognitive activity. Our findings provide insights about search behaviors and help inform the design of tools to support users working on multi-session search tasks.

KEYWORDS

Session Stopping Reasons, Cross-Session Search, Information Types, Cognitive Activity

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1 INTRODUCTION

When working on complex tasks, users often need to search across multiple sessions (referred to here as cross-session searches, or XSS) [23, 27]. This can occur for a variety of reasons. For example, the user may have found useful information that they need to try to apply to their task, or they may be fatigued from a long search session and need a break. Researchers have identified distinct search-stopping behaviors and analyzed them from diverse angles, including cognitive patterns, individual personalities, and search system features [6, 13, 15, 32, 39, 41]. Results from both theoretical and empirical studies suggest that motivations to stop a search can be broadly divided into two types: internal and external. Internal reasons originate from the searcher's own circumstances or feelings, such as having found all the necessary information, feeling satisfied with what has been found, or mental fatigue. Conversely, external reasons include distractions or unexpected interruptions, such as email alerts, incoming calls, or technical issues [13, 26, 28, 32, 41]. However, what remains less explored is how search-stopping reasons correlate with the information discovered and the cognitive journey of the searcher leading up to the decision to end the session.

The nature of information is a crucial factor in information behavior and can be affected by other factors involved in the information-seeking process. For example, prior researchers found that for simple tasks, people require more factual information like details, numbers, and addresses that can directly address their task questions. However, for complex tasks, searchers often need a greater variety of types of information, such as domain information for understanding task background; task-solving information with methods, strategies; and in-depth information to help make decisions and solve problems [10, 33, 42]. Additionally, people seek different types of information for different purposes as their task evolves. For instance, images can help creators to combine ideas, videos help them explore relevant information, and procedural information can help learn how to do things and plan out a project. [43, 44]. A goal of the present study is to examine if/how the types of information a searcher engages with during a search session influences why they stop that search session. Understanding this could aid search systems in supporting users' needs across different search sessions.

Another important aspect of search is the cognitive activities that a searcher engages in. The *search-as-learning* (SAL) community has identified the importance of cognitive activities involved in an information searching process. Researchers in SAL have explored how these activities relate to the complexity levels of the information tasks as well as to the information people discovered during the search process [12, 20, 21, 37].

Prior research in these areas motivated us to examine how searchers' cognitive activities and the information types they looked for relate to the information stopping behaviors. We conducted a diary study to investigate the cross-session search behaviors of people in real-life tasks. Over two months with twenty-five participants, we collected rich data about searchers' motivations for why they initiate/resume a search session and what made them decide to stop a session. Furthermore, our participants also provided detailed descriptions of the information they found and how they processed the information to achieve their task goals. The current paper focus on answering the following research questions:

- **RQ1:** What are the reasons that lead people to stop search sessions during the cross-session search process?
- **RQ2:** Whether and how do the session-stopping reasons relate to the types of information people found during the session?
- **RQ3:** Whether and how do the session-stopping reasons relate to the cognitive activities that people engaged in during the session?

2 RELATED WORK

Our work is built upon previous research on search-stopping reasons, cross-session searches, information types, and cognitive activities involved in the information-seeking process.

2.1 Search stopping reasons

Stopping behavior in Information Seeking (IS) and Information Retrieval (IR) has often been described as a point when a searcher finds the needed information or is satisfied with what they found and stops the ongoing search activities [19, 41]. Studies of search-stopping behavior often focus on two directions: modeling the search and stopping process, and investigating the factors that affect users' stopping behavior.

In 1968, Cooper [13] proposed two stopping rules: the "satisfaction stopping rule", where users stop after locating a set number of relevant documents, and the "frustration point rule", where users stop after finding too many unhelpful documents. Expanding on this, Kraft and Lee [24] introduced a hybrid approach where individuals stop upon reaching either satisfaction or frustration first. From a system perspective, both studies suggested that the search length could be predicted (expected search length) by considering the size of the database, the number of relevant documents, the number of relevant documents a searcher expected to find, and the number of irrelevant documents they could tolerate, etc.

Switching to a more cognitive, user-perspective approach, Nickles, Curley and Benson [32] found that people stop searching 1) when the search met a list of requirements they set earlier (mental list), 2) when their findings were sufficient and met a predetermined level (magnitude threshold), 3) when they cannot find any new information different from what they had saw (difference threshold), and 4) when they complete the construction of a mental model for the information relevant to their tasks (representational stability). Both the system-focused and cognition-focused stopping rules have been further investigated to decide whether they reflect real-world search-stopping behavior and which rules could be more efficient than others [5, 30].

Beyond static rules, researchers have identified other factors that explain search stopping behavior from diverse angles. For instance, Browne [5] noted a variance in strategies between expert and novice searchers. While experts often employed specific strategies such as a mental list, novices leaned more towards the difference threshold and representational stability. The nature of the task, such as task structure, can also affect users' search-stopping behavior and the underlying reasons. Dedema and Liu [15] found that participants applied mental list rules to stop specific queries at early stage of tasks that were well-structured and used threshold role to stop reading a webpage during the search process. In a recent comprehensive review, Ilani [19] classified the determinants of information search stopping behavior into two main categories: cognitive factors, which encompass searchers' individual traits like experience, skills, and emotions, and environmental factors, which cover elements such as the task's nature, the information retrieval system in use, and time constraints.

2.2 Cross-session search and stopping reasons

During cross-session search (XSS), people conduct a series of information seeking activities in multiple, distinct search sessions across a period of time. Researchers have investigated the characteristics of XSS from various aspects. For instance, researchers found that knowledge workers often were often involved in XSS to gather information from different sources [31, 34, 35]. Sellen et al. [34] and MacKay and Watters[28] found that information-gathering tasks were the most popular search task types involving XSS. On the contrary, Agichtein et al. [1]'s analysis of search log data from more than 1000 users suggested that information maintenance tasks and undirected browsing tasks were more likely to be continued across sessions than information gathering tasks. Researchers also found that the number of sessions and the time gap between sessions varies depending on the nature of the task and user characteristics [31, 35]. Recent studies related to XSS have explored multi-tasking and cross-device searching. For instance, Vuong et al.[38] found that users often conduct searches in parallel for different topics across various sources and applications. Wu et al.[40] and Han et al.[18] investigated users' cross-device search behavior when their participants sought for information across multiple sessions.

Researchers have also considered reasons that could cause users to stop a search session before they complete their overall task. Lin and Belkin[26] proposed the *Multiple Episode Information Seeking* (MISE) model which includes possible reasons that could cause people to stop and resume a search session as part of XSS. They outlined search stopping reasons such as external factors (e.g., time running out, distractions, mental or physical fatigue) and internal factors such as searchers' perceptions of the task situation, their task knowledge, and their cognition needs (e.g., to stop to process the found information). MacKay and Watters [28] observed students' XSS behavior and concluded that they often stopped because they found what they needed or their tasks were completed.

However, there is a paucity of studies to systematically investigate why people stopped search sessions when they are working on real-world tasks that involve XSS. Although Lin and Belkin's[26] MISE model indicated that users' information processing behavior could be a factor in causing session stopping, it was not clear

whether and how this behavior could affect the stopping decision during real-life XSS task process. We address this research question in the study presented in this paper.

2.3 Information types and cognitive activities during information search process

Information can be classified in various ways such as general versus specific, objective versus subjective, technical versus nontechnical, and factual versus opinionated [12]. The nature of a task can influence users' information needs for different types of information and their interaction with that information. Byström[8] defined three types of information used in professional decision-making tasks: 1) *task information*, referring factual data specific to a task, answering questions like names, address, and events. 2) *Domain information*, meaning background knowledge shared across tasks, including concepts, rules, theories, and known facts. 3) *Task-solving information*, meaning information about task structures and methods. It can encompass task or domain information but is primarily instructional.

Complex tasks demand a wider variety of information than simple tasks. For instance, simple tasks might only need task information, while complex ones might require all three types [9]. Saastamoinen et al. [33] found that simpler tasks favored isolated facts, while complex tasks leaned towards aggregated data. In health research, simpler tasks often utilize straightforward resources like FAQs, whereas complex tasks demand in-depth scholarly articles and reports [42].

In terms of cognitive activities, complex tasks often involve more uncertainty and require sophisticated cognitive activities, such as analyzing, evaluating, and planning [10, 33]. These cognitive functions during the information-seeking process are strongly linked to the learning process [7, 29].

Drawing from [2], researchers examined how task cognitive complexity influences users' search activities. For instance, Jansen et al. [20] observed distinct search behaviors for mid-level complexity tasks (applying and analyzing) compared to both lower (remembering, understanding) and higher-level tasks (evaluating, creating). In contrast, [22] found users spent more time and engaged more deeply with search systems for tasks with higher cognitive complexity. Recent studies, such as by [12], highlighted that tasks with higher cognitive complexity necessitate varied information types, especially subjective data. When provided with four information types (facts, concepts, opinions, insights), more complex tasks demanded a wider range of information types. Urgo et al. [37] incorporated both the cognitive complexity and knowledge type facets of Anderson and Krathwohl's taxonomy in their search task design. They found significant effects of information types on participants' perceptions of task difficulty, performance, and search patterns.

However, research remains sparse on the relationship between information types, cognitive activities during the search, and the reasons users decide to end their searches. XSS offers a unique lens to explore these relationships because: 1) as people progressively learn about their tasks in the XSS context, their information needs and cognitive activities may evolve, and 2) the XSS process, with its multiple search sessions, stopping and resuming, can shed light

on the evolving patterns of users' search activities and the reasons behind them.

3 METHOD

We conducted a diary study to investigate how and why people engage in XSS in the context of real-life tasks. The study was reviewed by the ethics board at our university.

3.1 Participant recruitment

We recruited participants via university-wide mailing lists. The recruitment email contained a screening questionnaire. The questionnaire outlined two requirements for eligibility: 1) the participant should have a real-world task or project they had yet to search for, and 2) the participant anticipated conducting multiple online searches to complete the task. Prospective participants were asked to provide a brief description of their tasks within the questionnaire. These prerequisites allowed us to focus our data collection on *anticipated* XSS tasks, which participants were already aware would require multiple search sessions. Based on this recruitment and screening, 29 people participated in the study.

3.2 Diary study process

During an introductory meeting with each participant, they completed a consent form and a pre-task questionnaire to provide more details about the task they planned to work on for the study. We also provided each participant with a personalized Google account to use for the study. This account came pre-configured with links to three online questionnaires: (1) a pre-session questionnaire, (2) a post-session questionnaire, and (3) a structured daily review diary. Additionally, a bookmark folder was set up allowing participants to save any web pages they deemed valuable for their tasks. Participants were then guided to access this account on their personal computers and perform a mock task. They were informed that their search history within this account would be recorded. However, they retained the right to review and erase any unrelated search history. To oversee the participants' workflow and to prompt them to fill out the required questionnaires and diaries, all the Google accounts were accessible by the study moderator. Participants were instructed to conduct all task-related searches using their designated Google account.

After the introductory meeting, participants worked on their tasks independently. Prior to initiating a search session related to their task, participants were directed to complete the pre-session questionnaire. Then, they could search however they wanted. They were also instructed that after they completed a search session, that they should fill out the post-session questionnaire. In addition, at the end of the post-session questionnaire, they were asked to provide up to three examples of information (url and text snippets) they found that helped them with their task. Furthermore, on any day they engaged with their task—be it searching or other related activities—they were asked to fill out the daily review diary.

Participants had the flexibility to conduct as many search sessions as they needed throughout the study. A minimum criterion for complete participation was the submission of data from at least three search sessions and two daily review diaries. Upon fulfilling the study's requirements, participants were compensated with

USD \$120 as compensation. Of the initial 29 participants, 15 were selected for a follow-up interview, for which they received an additional USD \$20. Analysis of the interview data is not included in the current paper.

3.3 Data collected and analysis

Our data collection spanned approximately two months, from August to October 2021. Four participants opted out of the study, hence their data was excluded. Of the remaining 25, there were: 9 undergraduates, 7 graduates, and 9 non-student staff from the university. Their age ranged from 18 to 62 years, with females comprising 60% of the group. The variety of tasks they completed was diverse, such as crafting a screenplay, assisting friends in finding an apartment to rent, planning a baby shower, initiating a small business, curating a LEGO collection, and writing an essay about an international problem. Collectively, these 25 participants submitted (a) 101 sets of completed pre/post-session questionnaires, (b) 279 examples of information found during their search sessions, and (c) 91 daily review diaries.

To address our **RQ1**, the post-session questionnaire included two questions: 1) an open-response question, "What are the reasons that made you stop this search session?", and 2) a single-selection multiple-choice question that asked participants to choose the *most important reason* that made them stop the session from the following list of options (based on stopping reasons suggested in Lin and Belkin's MISE model [26]):

- I needed to consult other sources (e.g., another person, organizations) for more information about the task
- I could not find the information I was looking for
- I needed to process/think about the gathered information before taking the next step
- I needed to validate the found information by applying it to my task.
- I found all the information I needed
- Task deadline was approaching.
- I could not complete the task.
- Other reasons (Please specify)

We used this two-question approach about why the participant stopped the session for two main reasons. First, the Lin and Belkin MISE stopping reasons [26] were based on a theoretical model, so we were not sure how well they would match participants' real-world stopping reasons. Second, we wanted to make sure that our understanding of why participants stopped a session matched the actual reasons why they stopped. By using two questions, we were able to be more confident in our qualitative coding of the stopping reason **and** were able to make sure that the participants' forced choice was consistent with the intended definitions of the stopping categories. This may seem like a lot of effort, but this mixed deductive-inductive approach revealed important insights and led to a final set of stopping reasons that differ from the original MISE categories.

In our qualitative analysis, we first coded the participant's response to the open-ended question and then checked this against their response to the single-selection question. In cases where there was a mismatch, we referred back to the open-ended description to gain a more complete understanding of the circumstances that

led to stopping the session and made a final determination. As presented in the results section, this process resulted in several of the MISE-based stopping reasons being merged/omitted and other reasons being added into the final results.

To address **RQ2** and **RQ3**, the post-session questionnaire requested that participants provide up to three examples of information they found during the session that was either helpful or not helpful. For each example, we asked them to: 1) describe the information found, 2) provide a URL, and 3) describe how they thought the information was helpful or not for their task. To analyze this data, we conducted deductive coding along two dimensions: the type of information, and the *cognitive processes (or activities)* that participants described in their responses of how the information was useful. To code these two dimensions, we used Anderson and Krathwohl's [3] taxonomy of learning (A&K). The A&K taxonomy consists of two dimensions, a *cognitive process* dimension and a *type of knowledge* dimension. These correspond to our cognitive activities and information types respectively. Our rationale for this approach stemmed from a previous study [25], where we found that most of the types of information users pursued during the XSS fit well with the A&K knowledge dimension and that their cognitive activities fit well within the A&K cognitive process dimension. Additional previous research has demonstrated that the A&K taxonomy is effective for investigating the nature of information and cognition during search processes [12, 37]. The original A&K taxonomy includes four knowledge types and six cognitive processes [3]. We adapted these slightly based on prior work in [12]. Our resulting coding scheme included four information types: factual, conceptual, procedural, opinion/ideas. For the cognitive activities, we used six categories: retrieving, understanding, applying, analyzing, evaluating, and creating.

For our coding of the cognitive activities that participants described in a session, we analyzed participants' descriptions of how they thought the information found was useful (or not) for their task. For our analyses, we selected the *highest-level* cognitive activity described (retrieving < understanding < applying < analyzing < evaluating < creating). We focused on analyzing the highest cognitive process described because the cognitive process levels build off each other (e.g. analyze may also requires understanding) and thus the highest levels of cognition represent the most complex cognitive behaviors of that search session.

A summary of the information type and cognitive level categories, definitions, and frequency counts (out of the 279 information examples in our data) are shown in Table 1.

4 RESULTS

Here we present our findings including: 1) the stopping reasons that caused participants to stop a search session during XSS; 2) different types of information people found useful to help them complete tasks; 3) the most complicated cognitive activities associated with the processed information during a search session.

4.1 Session stopping reasons during XSS

RQ1 asks "What are the reasons that lead people to stop search sessions during the cross-session search process?". To address this,

Table 1: Coding framework for information types and cognitive activity levels in XSS

Category	Type/Level	Definition (adopted from [3, 12, 37])	No.
Info. type	Factual	Fundamental details participants seek related to their task and the essential information needed for problem-solving	170
	Conceptual	Definitions and explanations of ideas, along with the interrelationships among various components within a topic, including principles, structures, rules, and standards	46
	Procedural	Guidance on 'how-to' execute a task, or information concerning the application of specific skills or techniques	39
	Opinion/ideas	Perspectives, remarks, and impressions expressed by individuals or groups.	24
Cog. level	Retrieving	Participants indicated they actively searched for and successfully located the information.	33
	Understanding	Participants actively integrate the retrieved information and connect it to their specific task context.	94
	Applying	Participants utilize the discovered information in specific scenarios or execute actions based on the provided details.	45
	Analyzing	Participants break down information into its components and discern how these elements interrelate and connect to the overarching problem.	18
	Evaluating	Participants assess the retrieved information against specific criteria and standards pertinent to their task.	76
	Creating	Participants synthesize the retrieved information, engaging in generation, planning, and design to produce a novel and cohesive result.	13

we analyzed data from two questions on the post-session questionnaire (as described in Section 3.3). Based on our qualitative analysis, we identified six primary reasons that led our participants to stop a search session at different stages of the XSS process: goal achieved, satisfied with the found information, interruption, need to consult other information sources, cannot find the needed information, low task knowledge (Table 2).

Table 2: Distribution of session stopping reasons at three task stages during XSS

Stopping Reason	Initial session	Mid-sessions(s)	Last session	Total
Goal achieved	8	15	11	34
Satisfied	10	14	10	34
Interruption	6	13	3	22
Need to consult others	1	4	1	6
Cannot find the needed info.	0	4	0	4
Low task knowledge	0	1	0	1
Total	25	51	25	101

We note that from the MISE proposed reasons, we did not observe any instances of "task deadline approaching" or "I could not complete the task", so there were omitted from our final list. We also observed that the MISE reasons "I needed to process/think about the gathered information" and "I needed to validate the found information" were often combined or the distinction was unclear in participants' descriptions, so we merged these two categories. Furthermore, "interruptions" was not included in the list of choices,

but many participants noted external interruptions as a reason for stopping a session in their open-ended responses.

Goal achieved. *Goal achieved* refers to situations when participants explicitly said that they achieved what they wanted for a session, often based on clear goals or quantified criteria. For example, p116 described, "I stopped because I completed my intentions for this first search." Overall, thirty-four ($n=34$) sessions were stopped because participants achieved their goals. Among them, eight ($n=8$, 24%) were initial sessions, fifteen ($n=15$, 44%) were middle sessions, and eleven ($n=11$, 32%) were the last session (Table 2).

Many of the *goal achieved* sessions were stopped because a participant completed a sub-goal or sub-stage of their overall main work task. For instance, p102 noted, "I am doing one session with one [sub-topic] a day, so after I finished I stopped." (p102-s4) Similarly, p126 commented, "I decided to stop because... [I] have reached a milestone." (p126-s3) As shown in Table 2, many of the last sessions ($n=11$) were stopped due to *goal achieved*. For example, "I stopped this search session because I found all the information I was looking for as well as guidance that I need to complete the [task]." (p116-s3) This example also illustrates that even though a searcher may have found all the information they needed, they might have had work to do to complete their overall *work task*.

We also note that *goal achieved* could happen for sessions that did not start with clear goals at the beginning. For instance, a participant may have only had a vague idea to explore first, but through the search process, they clarified what they needed and concentrated on these newly developed goals until they were achieved. For instance, participant (p107) described about the start of one of their sessions: "honestly, I had no idea what I was looking for" (p107-s3) By the end of the session, she said that "I... added all options... and then

dwindled down which would be best... A great successful task indeed!" (p107-s3)

Satisfied. *Satisfied* describes situations when people felt satisfied about the results they found by the end of a session. These cases often include expressions related to attitudes or content feelings (e.g., enough, good, satisfied, etc.) Sessions at any stages could be stopped because of *satisfied*: we found ten initial sessions (n=10, 29%), fourteen middle sessions (n=14, 41%), and ten last sessions (n=10, 29%).

Compared to sessions stopped because of *goal achieved*, the session goals of *satisfied* sessions were more general than specific (e.g., qualitative goal vs. quantitative goal). Participants might not have had ideas about what exact information they wanted, but as they explored and collected information online, they constructed mental models about their task and the needed information, and evaluated the information toward these goals till they felt good enough to stop. For instance, "I used this session to gather the bare minimum of information regarding my paper topic in order to familiarize myself with what I will be writing about over the next 2 weeks. I stopped this session because I found enough to feel comfortable with completing the task." (p113-s1)

We also found that *satisfied* was often associated with the expression of "natural stopping point" which refers to transit points where people could easily switch to another search topic or task stage in their next search session. For example, "It felt like something of a natural endpoint to the search session because I was at a good place to continue the next search more concretely" (p101-s1)

In addition, different from previous studies that have found that searchers need to stop search sessions to process the found information [26], we found that many of our participants felt satisfied after they cognitively processed the information during the search process and then decided to stop. For instance, "I synthesized this information to form a critical opinion that will support the thesis of my paper. I felt this was a good stopping point before continuing my research over the next few days" (p113-s4)

Interruption. *Interruption* describes reasons that are not related to users' information needs or task progress but that caused people to stop their current search session. In our data, 22 sessions out of the 101 that were stopped because of interruption. Interruptions could be self-initiated (e.g., mental/physical fatigue), or introduced by external events (e.g., phone calls, time commitment elsewhere). In our data almost all the interruptions were self-initiated, while only a few interruptions involved external factors/people. For instance, p107 noted, "It was getting late and I needed to sleep!" (p107-s3). Similarly, p124 described, "[I stopped because I need to] move locations from library to student union, as well as a low battery and need to charge laptop." (p124-s78)

Need to consult others. This stopping reason describes situations where participants stopped a search session because they wanted to consult with other people including both task collaborators and individuals regarded as experts in the field. In some cases, the people being consulted were directly involved in the overall work task. In these cases, consulting with the other people was often an integral part of moving the task forward. For instance, "I emailed [name]... I feel like I have learned a lot of information tonight... I can't officially settle on a dinner reservation until I hear

back from [name]." (p103-s3) In cases where the people being consulted were not part of the overall work task, participants often thought those people could provide additional information (e.g., expertise, similar experiences) that could help them know more about the task. For instance, "I also wanted to get my friends' opinions..." (p101-s1). In some cases, the other information being consulted was an information source that they could not access during the search session (or needed extra assistance to access), such as a physical library resource. For instance, "I also realized the source I wanted to use for further source finding is one I'll have to get physically through interlibrary loan." (p102-s1)

Cannot find the needed information. This stopping reason refers to situations when participants: 1) have an idea about what information they want, and 2) spent a significant amount of time and effort looking for information to meet their needs, and 3) did not find what they were looking for and decided to stop. In our data, there were four (n=4) sessions where participants stopped because they could not find what they needed. For instance, p103 noted, "I was mildly annoyed that I couldn't find a simple cocktail recipe and also frustrated that the cake websites didn't have an allergies section." (p103-s2). And p106 described, "I'm too frustrated with inconsistent information and websites of shuttle companies that don't work. I think I'm going to just take a Lyft or Uber for simplicity and not have to deal with booking a shuttle in advance." (p106-s3) We observed several different reasons that people were not able to find what they needed: (1) the needed information was not provided or available; (2) the provided information did not work, or (3) the participant did not find information that matches their requirements.

It is interesting to note that during XSS, people have opportunities to resume their previous searches. However, we found that for sessions that were stopped because of *cannot find the needed information*, often our participants did not resume searching for the same information during later sessions. Instead, participants often looked for alternative choices or just stopped searching for the related sub-topics. For instance, "I hit a dead end in my search. I wasn't finding much else, and I wasn't super interested in looking for busted [name] since most of them weren't ideal colors and I don't want to paint them. Nonetheless, I found enough options to be able to choose what I need later." (p118-s2)

Low task knowledge. Previous research [26] has noted that a user having a low understanding of the goal could also cause the session to be interrupted (i.e., internal factors). In our data, we found one example where a participant stopped a session because they felt overwhelmed. P120 noted, "I felt overwhelmed by my uncertainty about going too far in one direction and eventually uninspired to continue." (p120-s3). We speculate that the reason why *low task knowledge* was not a common stopping reason may be that during XSS, participants have opportunities to explore the topic from many different dimensions. Thus, they could incrementally gain knowledge about their task topic, even if it was unfamiliar to them at the start of their search process.

4.2 Stopping reasons and information types

In this section, we consider whether the information types found had a relationship with the stopping reason (RQ2). Recall that participants submitted up to three examples of information that they found during each search session. Across the 101 search sessions, we collected 279 information examples which we classified using four information types (factual, conceptual, procedure, opinion).

Table 3 summarizes the frequencies of different information type examples across the session stopping reasons. For example, participants submitted a total of 60 *factual* information examples for search sessions that were stopped because of *goal achieved*.

Table 3: Number of information type examples for sessions stopped by different reasons

Stopping/InfoType	Facutal	Conceptual	Procedural	Opinion	Total
Goal achieved	60	8	10	14	92
Satisfied	61	22	9	7	99
Interruption	25	16	17	3	61
Consult	16	0	1	0	17
Cannot find	6	0	1	0	7
Low task know.	2	0	1	0	3
Total					279

Figure 1 shows the percentages of the different information types for each session stopping reason.

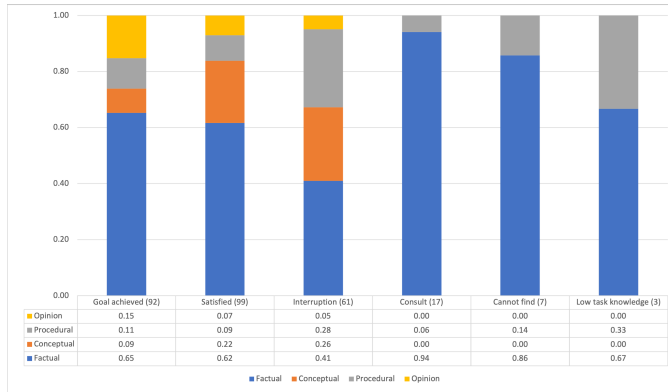


Figure 1: The percentage of information types for sessions stopped for different reasons.

Across all the information types and stopping reasons, a chi-square test found a significant association between the reasons why people decided to stop a session and the information types they found during the session ($\chi^2(15,279)=41.97, p < 0.001$). However, with six stopping reasons and four information types, additional analysis was needed to understand the association. To gain insights, we decided to drop the *consult*, *cannot find*, and *low task knowledge* stopping reasons since they had low frequencies. Using just the *goal achieved*, *satisfied*, and *interruption* reasons, a chi-square test still found a significant association ($\chi^2(6,252)=27.29, p < 0.001$). Next,

we focused on differences between two pairs of stopping reasons: 1) goal achieved vs. interruption, and 2) goal achieved vs. satisfied.

Goal Achieved vs. Interruption – Visual inspection of Figure 1 shows that interruption had a lower frequency of factual information and larger frequencies of procedural and conceptual information compared to goal achieved. A chi-square between goal achieved and interruption was significant ($\chi^2(3,153)=20.57, p < 0.001$), supporting association. One possible reason for this could be that engaging with procedural and conceptual information may require more time (compared to factual information), which could increase the chances of being interrupted.

Goal Achieved vs. Satisfied – Inspection of Figure 1 shows that factual and conceptual are two frequent information types in the goal achieved and satisfied categories. A chi-square between goal achieved and satisfied was significant ($\chi^2(3,191)=8.68, p < 0.05$), indicating an association. In addition, a chi-square test with stopping reasons (goal achieved and satisfied) and information types (factual and conceptual) was significant ($\chi^2(1,151)=5.10, p < 0.05$), showing that the goal achieved stopping reason had fewer conceptual information types than satisfied. A possible explanation for this is that for *goal achieved* sessions, participants often had clear information goals or criteria, whereas for *satisfied* sessions their goals were more general. Thus, the goal achieved sessions may not require as much broad, conceptual information. Similarly, there is a trend that goal achieved sessions involved more opinion information, compared to satisfied sessions. This may suggest that opinion information is more common/useful when a person is working toward specific information goals.

4.3 Stopping reasons and cognitive activities

RQ3 explores whether and how session stopping reasons relation to cognitive activities that people engaged in during their search session.

Table 4: Number of different stopping reasons for all sessions represented by their most complicated cognitive activity

Stopping/Cog.	R	U	App.	Ana.	E	C	Total
Goal achieved	1	4	4	7	16	2	
Satisfied	0	8	5	2	15	4	
Interruption	0	7	2	0	7	6	
Consult	0	0	0	0	6	0	
Cannot find	0	0	1	0	2	0	
Low task know.	0	1	0	0	0	0	
Total	1	20	12	9	46	12	100

Table 4 shows the number of sessions that reached each cognitive activity level that were stopped for each stopping reason. We note two important points about Table 4. First, the data shows the *highest* level of cognitive activity reached during the session. For example, during the 46 sessions in which participants reached the evaluate cognitive level, they may have also engaged in lower-level cognitive activities (e.g., understanding, analyzing). Second, we note that the analysis in this section is at the *session* level (i.e. the 101 sessions that our participants reported). One session is left out of this analysis

because the participant reported that they found nothing for that session, so the totals in Table 4 are out of 100 sessions.

To investigate RQ3, we ran chi-square tests to look at the association between session stopping reasons and the highest level of cognitive activity reached during the session. Across all the data, a chi-square test did not find a significant association ($\chi^2(25,100)=32.58, p=0.142$). Because data for some stopping reasons and cognitive levels were sparse (see the left column and lower rows of Table 4), we ran additional chi-square tests on subsets of the categories to see if there might be associations among the variables for which we had sufficient data. Specifically, we conducted two additional chi-squared analyses: 1) a chi-square with three stopping reasons (goal achieved, satisfied, and interruption) and with all the cognitive activities except remember ($\chi^2(8,89)=15.12, p=0.057$); and 2) a chi-square with two stopping reasons (goal achieved, satisfied) and two cognitive activities (understand, evaluate) ($\chi^2(1,43)=1.16, p=0.281$). Neither was significant. Thus, we report that we did not find a significant association between stopping reasons and cognitive activities.

However, we note several important trends that are evident in our data. First, as shown in Table 4, 46 out of the 100 sessions reached the cognitive level of *evaluate*. It is interesting that such a large percentage of the overall sessions reached the *evaluate* cognitive level. This suggests if a search system detects that a person is engaged in a cross-session search task, they may benefit from features that assist them with evaluate-level activities (e.g. comparison, decision-making, etc). Second, Table 4 shows that 20 out of the 100 sessions reached the *understand* cognitive level. This suggests that a large percentage of searchers doing XSS might benefit from tools to aid their understanding of the topics they are searching about.

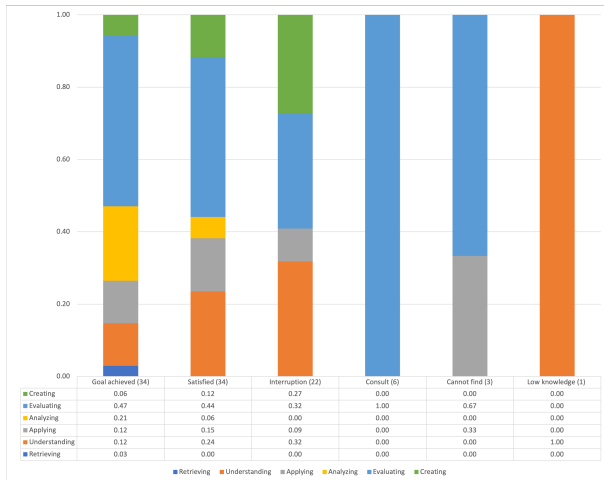


Figure 2: The percentage of sessions that stopped for different reasons over the sessions that involved a specific type of cognitive activity as the most complicated cognitive activity.

Finally, we note that Figure 2 shows that all the sessions that were stopped to "consult with others" reached the *evaluate* cognitive level. At the evaluate level, people need to assess information

against specific criteria and are often trying to reach a decision about how to proceed. One possible explanation for this could be that when people stopped a session to consult with others, they had already progressed through the lower-level cognitive activities such as retrieving and understanding. In other words, when they stopped a session to consult with others, they had reached a high-level cognitive activity (evaluate) that required additional input or information from others.

5 DISCUSSION AND IMPLICATION

In this paper, we present session stopping reasons found during participants' real-life cross-session search (XSS) processes. We also present results about the relationships of these stopping reasons with the information types people found useful for their tasks and the representative cognitive activities regarding how people processed the information during the corresponding sessions.

For RQ1, we identified six primary reasons that caused people to stop a search session for tasks involving XSS: goals achieved, satisfaction, interruption, cannot find the needed information, to consult other information sources, and low task knowledge. Among them, goals achieved, satisfaction and interruptions were more common than other reasons. This is consistent with previous studies that found that 1) searchers often stopped their searches because they found what they needed or they felt "satisfied" with what they found [28, 41], and 2) that complex information tasks are often interrupted and require resumptions [31].

However, different from previous research that considered stopping reasons in single session searches, or towards the end of a search task, our findings indicate that people may stop search sessions before they "complete" their overall goals or tasks. People may stop because they achieved sub-goals of a task, because they are satisfied with what they found so far, or because they need to consult with others. However, instead of walking away from the task, they expected to come back and continue their search till they accomplished their entire task. Urgo and Arguello [36] conducted an empirical study and found that helping participants set up sub-goals can improve searchers' task performance and could help them resume the tasks when they come back later. Our study supports this finding. In real-life examples with complex tasks, people often expect to search across multiple sessions and intentionally stage goals for different sessions as they complete their overall task.

In addition, by starting with Lin and Belkin's theoretical model [26], our findings modify and extend the MISE reasons with the three session stopping reasons that were less mentioned in previous studies: stop to consult other sources, cannot find the needed information, and low task knowledge. As earlier studies treated these reasons as barriers that may cause search failure [39], our findings indicate that these stopping reasons in XSS may lead to a participant coming back and searching again (perhaps in a new direction). Again, stopping is not finishing. Future search systems may consider detecting these reasons to provide more customized information: information from more sources or in different formats for situations when people need to consult other information sources, search intervention by involving more advanced search assistance like generative AI or crowdsourcing opinions from experts, breaking down the search topics into more manageable levels to help clarify the information

needs, etc. Last but not least, modeling for task continuation predicting could also take the stopping reasons as a factor, especially for tasks involving multiple search sessions.

For **RQ2**, our findings show that there were statistically significant relationships between the session stopping reasons and the information types, indicating that the information types encountered may be related to the reasons why people stopped their sessions during the XSS process. It is unsurprising that factual information is the most common information type people looked for across all the sessions. However, our results show that when sessions involved factual information, they were less often stopped because of interruption than for other reasons. This might be because, compared to other information types, factual information is about specific details that can be addressed by single query search or can be found from single information sources without cross-referring multiple places. Since searching for these types of information is fast and efficient using modern search engines, interruptions for this type of search may occur less frequently.

On the contrary, in our data, sessions that involved procedural information were frequently stopped due to interruptions. This suggests that sessions that involve searching for procedural information might take more time to accomplish, creating more opportunities for interruptions. Previous research found that searching for procedural information is ubiquitous, but that it involves many constraints [4, 11]. For instance, Ertl [16] found that understanding procedural information requires more prior knowledge of relevant concepts. As a result, searching for procedural information could be interrupted because of the searcher's lack of background knowledge of relevant concepts. In another study [17], researchers found that when searching for procedural information, searchers were often challenged by information overload, inaccurate information, and too broad search results, which could lead to interruptions (e.g., due to fatigue). In addition, [11] revealed that users are often involved in a "trial-and-error" pattern to use procedural information to solve problems, which again supports our finding that when procedural information sessions stopped, people may not complete their task yet but just need try what they found to decide next step.

For **RQ3**, we found no statistically significant relationship between the reasons for stopping a session and the most complex cognitive activities undertaken during those sessions. However, we did observe that almost half the sessions reached the *evaluate* cognitive complexity level. This is an important empirical finding that extends prior work. For example, prior research has found that users often engage in XSS to gather information from different sources [28, 31, 34, 35]. Our findings show that cross-session searches are not only about information gathering (retrieval), but also commonly involve higher-level cognitive activities. Here, we find that not only were the over-arching work tasks of our participants complex in nature, but also that a large number of the individual search sessions in support of those tasks led users to high-levels of cognitive activity.

Prior researchers introduced search stopping reasons related to cognition either focusing on "when" people decide to stop based on pre-determined rules [32] or proposing that people stopped searching to cognitively process the found information separately Lin and Belkin's [26] MISE model. Our findings extend the research by fully considering how people process the information as they

search. The trends we observe provide insights into future study directions such as to achieving certain stopping rules (e.g., mental list, different threshold), what types of cognitive activities are more common in XSS requires external assistance. We found evaluating was frequently found as the most complicated cognitive activity for sessions stopped for most of the reasons except the one stopped for low task knowledge. As being mentioned earlier, this could indicate that 1) many complex tasks involve the evaluating process to determine whether the information found was appropriate to the tasks at hand, 2) as the search process spreads over time, participants always need to evaluate whether they have collected enough information and when they could stop searching for more. As the activity of evaluating information retrieved require knowledge about the task and needed information; the activity of evaluating the enoughness of the information towards task completion requires metacognition skills like planning and monitoring. This provides insights for developing search assistant tools to support both cognitive evaluating skills (i.e., comparing, differencing, assessing) and metacognitive strategies including planning, monitoring of one search strategies and task process. This echos previous call for tools to support complex cognitive and metacognitive activities during exploratory search [14]. Future design endeavors might consider to explore interface designs that provide users with feedback about their search progress and strategy efficacy, empowering them to make more informed decisions during their search sessions. Given the apparent disconnect between cognitive activities and stopping reasons, there might be value in developing training modules that guide users on how to optimally process and evaluate information during their searches.

Limitations: Our study has several limitations. First, there is a that participants' answers to the single choice question and to the open-ended questions about their session stopping reasons might have influenced each other, especially since participants answered these questions for each search session. Second, the generalizability of the results could be limited. Although our participants had diverse backgrounds and brought their own tasks to our study, all participants were recruited from our university which may not accurately reflect broader populations or different kinds of tasks. Lastly, to explore the relationships between the session stopping reasons and other task factors, we conducted a series of comparisons using chi-square tests. Given that we did not control the task types or the search process for this diary study, the data we collected was not balanced or representative of all combinations (i.e., some rows and columns of Table 4 were sparse). Such kind of diversity in the tasks can make it challenging to draw general conclusions. To better understand these relationships, further experimental studies in a more controlled setting are recommended.

6 CONCLUSION

In this paper, we present our findings on the reasons that cause people to stop a search session and its relationships with 1) the information types people encountered and 2) the most complicated cognitive activities regarding how they processed the found information in the context of tasks involving XSS.

We identified six primary session stopping reasons, including goal achieved, satisfied, interruption, to consult others, cannot find

the needed information, and lack of task knowledge. Through a series of Chi-square tests, we found that the session stopping reasons are significantly associated with the information types (i.e., factual, conceptual, procedural, opinion/ideas) people found during the sessions. We did not find significant associations between the stopping reasons and the most complicated cognitive activities undertaken. However, we did observe that almost half the search sessions reached the *evaluate* cognitive level, indicating that many sessions in XSS are cognitively demanding.

Our results indicate that within the XSS context, there appears to be an association between the types of information encountered and the session stopping reasons. Furthermore, our results suggest that many of the individual sessions in a cross-session search reach high levels of cognitive activity. Our findings inform the design of future search systems to predict session stopping reasons based on the types of information they interact with during the search process. Furthermore, the observed cognitive patterns during XSS indicate that enhancing search systems isn't merely about surfacing the most relevant results. There's evident need for tools that facilitate users in cognitively evaluating the information and in performing metacognitive assessment of both their task's progression and their search strategies.

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